7. Variational Autoencoder

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FS26 MTH345: Deep Generative Models: A Statistical Perspective





Lecture Topics

M1. Foundations of Probabilistic Modelling

1. Fundamentals of Probability Theory
2. Random Variables
3. Likelihood Maximization Principle
4. Basics of Probabilistic Modelling

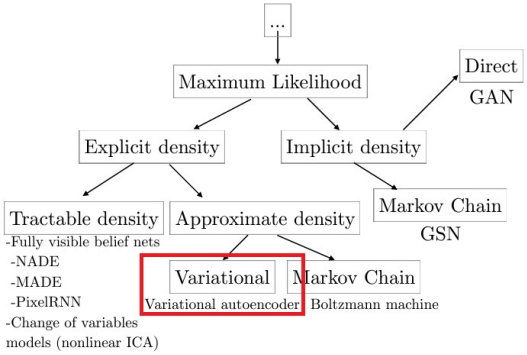
M2. Likelihood Ratio-based Generative Models

5. Autoregressive Model
6. Energy-based Model
7. Variational Autoencoder
8. Generative Adversarial Network

M3. Generative Models with Other Statistical Distance

9. Integral Probability Metric-based Approaches
10. Wasserstein Distance-based Approaches
11. Score-based Approaches

Lecture Topics



- In the previous chapter, we reviewed energy-based models. Both training and generation require running a Markov chain (with multi-step Gibbs sampling).
- In this lecture, we will review another generative model, *variational autoencoders (VAEs)*. By applying variational inference, they do not require running Markov chains during either training or testing times.

The figure is edited from Goodfellow (2016).

Lecture Topics

- Recommended reference materials include:
  - Kingma and Welling (2014), "Auto-encoding variational Bayes," *Proceedings of the International Conference on Learning Representations (ICLR)*.
  - Kingma et al. (2019), "An Introduction to Variational Autoencoders," *Foundations and Trends® in Machine Learning*.
  - Goodfellow (2016), "Nips 2016 tutorial: Generative adversarial networks," *arXiv*.

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Motivation

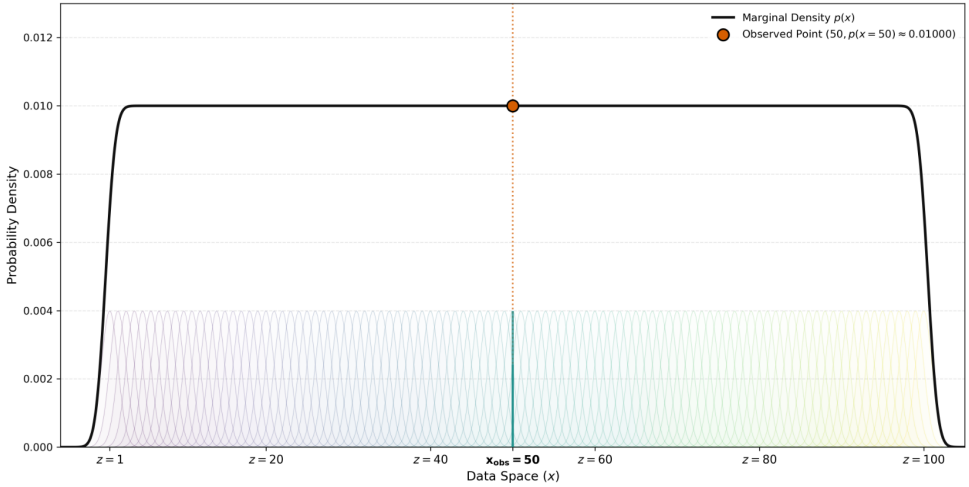
- As a preliminary, we briefly review the framework of *variational inference* (Bishop, 2006).
- Consider a discrete mixture model with a latent variable $Z \in \{1, \ldots, K\}$ and an observed variable X. The generative process is a Gaussian mixture model and is assumed to be known:[1]

$$\begin{aligned} \boldsymbol{Z} &\sim \text{Multin}(1, (1/K, \ldots, 1/K)^T) \\ X \mid \boldsymbol{Z} = \boldsymbol{e}_k &\sim \mathcal{N}(k, 1). \end{aligned} \tag{1}$$

- Evaluating the marginal likelihood $p(x)$ poses a severe computational challenge, and variational inference provides a prominent framework to circumvent this issue.

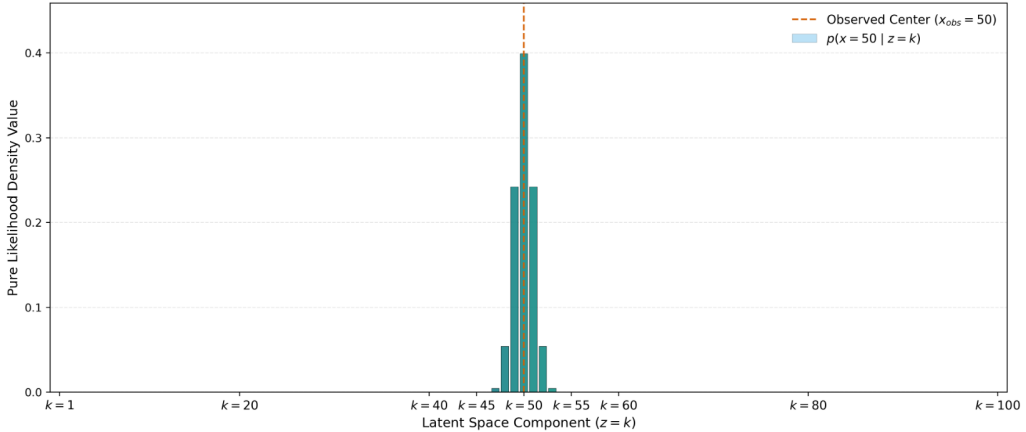
[1] From now on, following the convention in variational inference, we use p to denote the PDF regardless of the data type.

Motivation



- The $p(x)$ can be expressed as
  $p(x) = \sum_{k=1}^{K} p(x \mid \boldsymbol{z} = \boldsymbol{e}_k) p(\boldsymbol{z} = \boldsymbol{e}_k) = \sum_{k=1}^{K} \phi(x-k)/K.$

Motivation



- Most of the values $p(x \mid \boldsymbol{z} = \boldsymbol{e}_k)$ are extremely close to zero, except within the narrow interval $k \in \{47, \ldots, 53\}$.

Naive Monte Carlo

- When K is large or when Z is continuous, the analytical or numerical computation of the marginal likelihood $p(x)$ is often intractable.
- Here, we first consider an intuitive baseline approach.
- **Naive Monte Carlo**: Draw $z_{\text{naive}}^{(1)}, \ldots, z_{\text{naive}}^{(M)} \overset{\text{i.i.d.}}{\sim} p(z)$ and calculate:

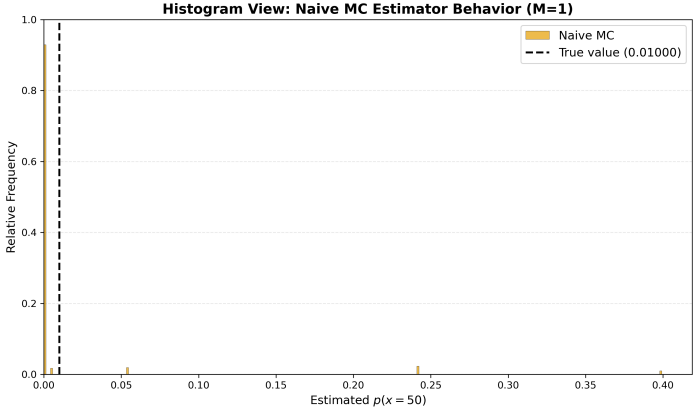
$$\hat{p}_{\text{naive}}(x) := \frac{1}{M} \sum_{m=1}^{M} p\big(x \mid z_{\text{naive}}^{(m)}\big). \tag{2}$$

- For example, when estimating $p(x = 50)$ with $M = 3$, if the drawn samples are $\boldsymbol{z}_{\text{naive}}^{(1)} = \boldsymbol{e}_{12}$, $\boldsymbol{z}_{\text{naive}}^{(2)} = \boldsymbol{e}_{50}$, and $\boldsymbol{z}_{\text{naive}}^{(3)} = \boldsymbol{e}_{84}$, the empirical calculation yields:

$$\hat{p}_{\text{naive}}(x = 50) = \frac{0 + \frac{1}{\sqrt{2\pi}} + 0}{3} = \frac{1}{3\sqrt{2\pi}} \approx 0.13298. \tag{3}$$

Recall that the true marginal likelihood is
$p(x = 50) = \frac{1}{100} \sum_{k=1}^{100} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(50-k)^2}{2}\right) \approx 0.01000.$

Naive Monte Carlo



- The empirical calculation results heavily depend on the samples falling to $\{47, \ldots, 53\}$. Nevertheless, the Monte Carlo estimator is unbiased.

Naive Monte Carlo

Exercise 1

Let $Z^{(1)}, \ldots, Z^{(M)} \overset{\text{i.i.d.}}{\sim} p(z)$, and define the Monte Carlo estimator of the marginal likelihood as:

$$\hat{p}_{\text{naive}}(x) := \frac{1}{M} \sum_{m=1}^{M} p\left(x \mid Z^{(m)}\right). \tag{4}$$

Show that $\hat{p}_{\text{naive}}(x)$ is an unbiased estimator of $p(x)$, i.e.,

$$\mathrm{E}\left[\hat{p}_{\text{naive}}(x)\right] = p(x). \tag{5}$$

Naive Monte Carlo



Naive Monte Carlo

- Its high variability can be expressed as:

$$\begin{aligned} \mathrm{Var}_{Z_{\text{naive}}^{(m)} \overset{\text{i.i.d.}}{\sim} p(z)} \left[\hat{p}_{\text{naive}}(x) \right] &= \frac{1}{M} \mathrm{Var}_{Z \sim p(z)} \left[p(x \mid Z) \right] \\ &= \frac{1}{M} \left(\mathrm{E}_{Z \sim p(z)} \left[p(x \mid Z)^2 \right] - p(x)^2 \right). \end{aligned} \tag{6}$$

- By applying the CLT, as $M \to \infty$,

$$\begin{aligned} \sqrt{M} \left(\hat{p}_{\text{naive}}(x) - p(x) \right) \xrightarrow{d} & \mathcal{N} \left(0, \mathrm{Var}_{Z \sim p(z)} \left[p(x \mid Z) \right] \right) \\ = & \mathcal{N} \left(0, \mathrm{E}_{Z \sim p(z)} \left[p(x \mid Z)^2 \right] - p(x)^2 \right). \end{aligned} \tag{7}$$

Importance Sampling

- One main source of the high variability of the Naive Monte Carlo approach is that the prior $p(z = k)$ does not reflect the relative contribution of $p(x \mid z = k)$ to the marginal likelihood $p(x)$.
- **Importance Sampling**: Draw $z_{\text{is}}^{(1)}, \ldots, z_{\text{is}}^{(M)} \overset{\text{i.i.d.}}{\sim} q(z)$ and calculate:

$$\hat{p}_{\text{is}}(x) := \sum_{m=1}^{M} p\big(x \mid z_{\text{is}}^{(m)}\big) \frac{p\big(z_{\text{is}}^{(m)}\big)}{M q\big(z_{\text{is}}^{(m)}\big)}. \tag{8}$$

  The $q(z)$ is referred to as the *proposal distribution*.
- By appropriately choosing $q(z)$ to yield high probability values where the joint density $p(x, z) = p(x \mid z)p(z)$ is significant, we can expect to reduce the variability of the estimator.

Importance Sampling

Exercise 2

Let $Z^{(1)}, \ldots, Z^{(M)} \overset{\text{i.i.d.}}{\sim} q(z)$, and define the importance sampling estimator of the marginal likelihood as:

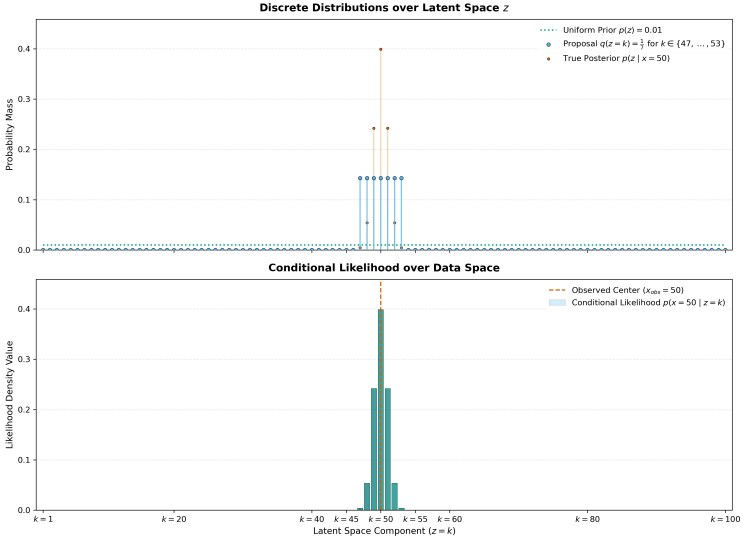
$$\hat{p}_{\text{is}}(x) := \frac{1}{M} \sum_{m=1}^{M} p\big(x \mid Z^{(m)}\big) \frac{p\big(Z^{(m)}\big)}{q\big(Z^{(m)}\big)}. \tag{9}$$

Show that $\hat{p}_{\text{is}}(x)$ is an unbiased estimator of $p(x)$, i.e.,

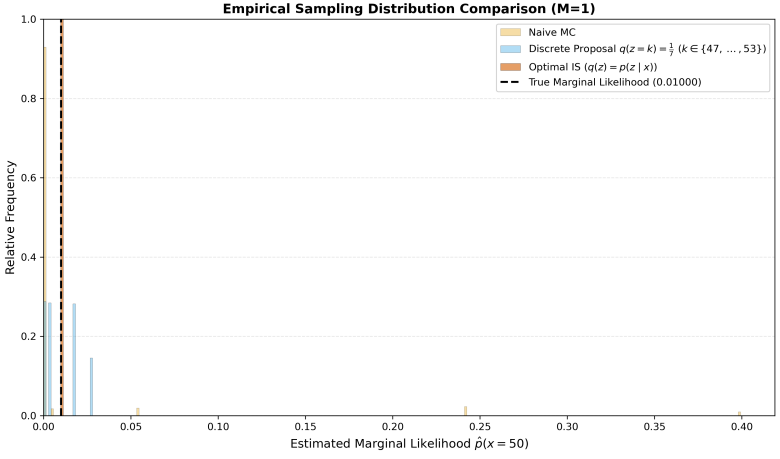
$$\mathrm{E}\left[\hat{p}_{\text{is}}(x)\right] = p(x). \tag{10}$$

Importance Sampling

Importance Sampling



Importance Sampling



Importance Sampling

- The variance of the importance sampling estimator can be expressed as:

$$\begin{aligned}\mathrm{Var}_{Z_{\mathrm{is}}^{(m)} \overset{\text{i.i.d.}}{\sim} q(z)}\left[\hat{p}_{\mathrm{is}}(x)\right] &= \frac{1}{M}\,\mathrm{Var}_{Z\sim q(z)}\left[\frac{p(Z,x)}{q(Z)}\right] \\ &= \frac{1}{M}\left(\mathrm{E}_{Z\sim q(z)}\left[\left(\frac{p(Z,x)}{q(Z)}\right)^2\right] - p(x)^2\right).\end{aligned} \tag{11}$$

- By applying the CLT, we have:

$$\begin{aligned}\sqrt{M}\left(\hat{p}_{\mathrm{is}}(x) - p(x)\right) &\xrightarrow{d} \mathcal{N}\left(0,\, \mathrm{Var}_{Z\sim q(z)}\left[\frac{p(Z,x)}{q(Z)}\right]\right) \\ &= \mathcal{N}\left(0,\, \mathrm{E}_{Z\sim q(z)}\left[\left(\frac{p(Z,x)}{q(Z)}\right)^2\right] - p(x)^2\right).\end{aligned} \tag{12}$$

Importance Sampling

Exercise 3

Let $Z^{(1)}, \ldots, Z^{(M)} \overset{\text{i.i.d.}}{\sim} q(z)$, and define the importance sampling estimator of the marginal likelihood as:

$$\hat{p}_{\text{is}}(x) := \frac{1}{M} \sum_{m=1}^{M} p\big(x \mid Z^{(m)}\big) \frac{p\big(Z^{(m)}\big)}{q\big(Z^{(m)}\big)}. \tag{13}$$

Show that $\text{Var}\left[\hat{p}_{\text{is}}(x)\right] = 0$ if and only if $q(z) = p(z \mid x)$.

Variational Inference

- The ideal choice for $q(z)$, the true posterior $q^*(z) = p(z \mid x)$, is often computationally intractable.
- **Variational Inference**: We can reformulate the intractable inference problem into a deterministic optimization framework:

$$q_{\phi^*} = \arg \min_{q_\phi(z) \in \mathcal{Q}} \mathcal{D}_{\mathrm{KL}}\big(q_\phi(z) \parallel p(z \mid x)\big), \tag{14}$$

where $\mathcal{Q}$ denotes a tractable family of distributions, a paradigm also known as *Variational Bayes*.

Variational Inference

- Since q_{ϕ^*} depends on the observed data x, we map each observation to its corresponding data-dependent proposal:

$$x \mapsto q_{\phi^*}(z \mid x) := \arg \min_{q_\phi(z|x) \in \mathcal{Q}} \mathcal{D}_{\mathrm{KL}}\big(q_\phi(z \mid x) \parallel p(z \mid x)\big). \tag{15}$$

- The *Variational Autoencoder (VAE)* operationalizes this generalized encoding process by parameterizing the mapping via a neural network.

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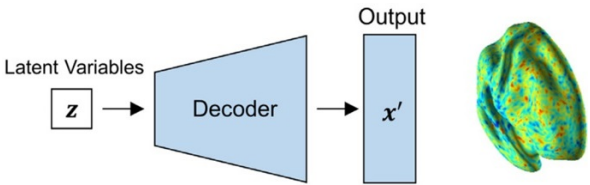
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4. Temporal Variational Autoencoder

Generation with Nonlinear Decoder



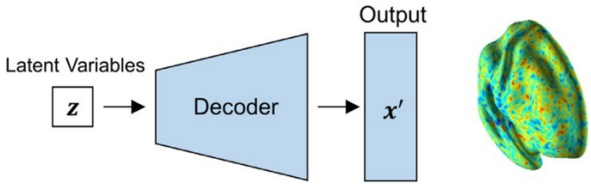
- How about generating data using a decoder? That is, the density is modeled as

$$p_\theta(\boldsymbol{z}, \boldsymbol{x}) := p(\boldsymbol{z}) p_\theta(\boldsymbol{x} \mid \boldsymbol{z}), \tag{16}$$

where $p(\boldsymbol{z})$ be prior (e.g., Gaussian) and $p_\theta(\boldsymbol{x}|\boldsymbol{z}) = \mathcal{N}(\boldsymbol{x}; \boldsymbol{\mu}_\theta(\boldsymbol{z}), \Sigma_\theta(\boldsymbol{z}))$ with $\boldsymbol{\mu}_\theta(\cdot)$ and $\Sigma_\theta(\cdot)$ being neural networks parameterized by θ.

Images are edited from Kim et al. (2021).

Generation with Nonlinear Decoder



- This approach extends the decoder (μ_θ) to a probabilistic model. When we sample $\boldsymbol{Z}$ multiple times from $p(\boldsymbol{z})$ and sample $\boldsymbol{X}$ from this (conditional) Gaussian, the distribution of generated $\boldsymbol{X}$ would be: $p_\theta(\boldsymbol{x}) = \int p(\boldsymbol{z}) p_\theta(\boldsymbol{x}|\boldsymbol{z}) d\boldsymbol{z}$.
- Due to the flexibility of nonlinear decoder networks, finding the MLE becomes highly challenging:

$$\theta^* = \arg\max_\theta \sum_{i=1}^{n} \log p_\theta(\boldsymbol{x}_i). \tag{17}$$

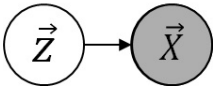
Images are edited from Kim et al. (2021).

Generation with Nonlinear Decoder

Exercise 4

Let $\boldsymbol{Z} \sim \mathcal{N}(0, I_r)$ and $\boldsymbol{X} \mid \boldsymbol{Z} = \boldsymbol{z} \sim \mathcal{N}(\boldsymbol{\mu}_\theta(\boldsymbol{z}), \Sigma_\theta(\boldsymbol{z}))$. Express the marginal density $p_\theta(\boldsymbol{x})$.

Variational Autoencoder

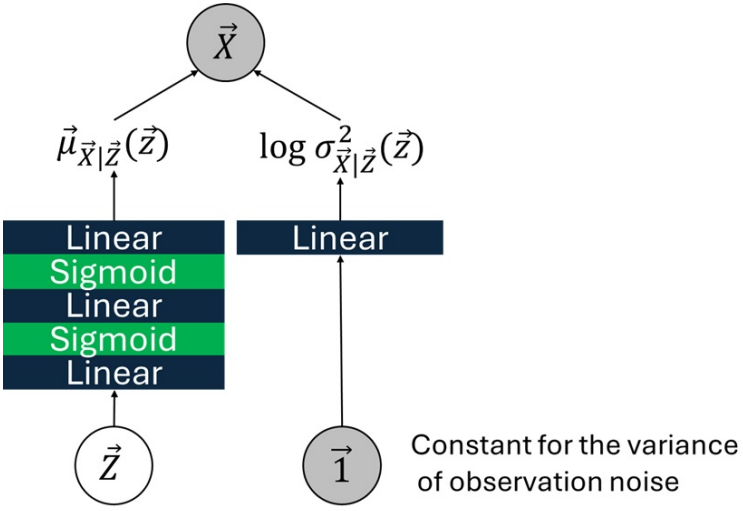


- Variational Autoencoders (VAEs, Kingma and Welling, 2014) utilize variational inference to overcome the aforementioned challenges.
- They model the data generation process using decoder networks with a diagonal covariance $\Sigma_\theta := \text{diag}(\sigma_1^2, \ldots, \sigma_d^2)$:

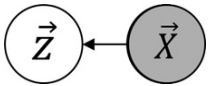
$$p_\theta(\boldsymbol{z}, \boldsymbol{x}) = \Big(\prod_{j=1}^{r} p(z_j)\Big)\Big(\prod_{j=1}^{d} p_\theta(x_j|\boldsymbol{z})\Big)$$

$$= \prod_{j=1}^{r}\left(\frac{1}{\sqrt{2\pi}}\exp\left(-\frac{z_j^2}{2}\right)\right)\prod_{j=1}^{d}\left(\frac{1}{\sqrt{2\pi\sigma_j^2}}\exp\left(-\frac{(x_j - \big(\mu_\theta(\boldsymbol{z})\big)_j)^2}{2\sigma_j^2}\right)\right). \quad (18)$$

Variational Autoencoder



Variational Autoencoder



- VAEs introduce an encoder, or *inference network*, $q_\phi(\boldsymbol{z}|\boldsymbol{x})$ to approximate $p_\theta(\boldsymbol{z}|\boldsymbol{x})$.[2]
- By applying the variational inference, we have:

$$\log p_\theta(\boldsymbol{x}) = \log \mathrm{E}_{\boldsymbol{Z}\sim q_\phi(\boldsymbol{z}|\boldsymbol{x})}\left[\frac{p_\theta(\boldsymbol{z},\boldsymbol{x})}{q_\phi(\boldsymbol{z}\mid\boldsymbol{x})}\right]. \tag{19}$$

[2]VAEs use $q_\phi(\boldsymbol{z}|\boldsymbol{x})$, a single function (of $\boldsymbol{x}$), to approximate the posterior distribution for each new data point, referred to as *amortized variational inference*.

Variational Autoencoder

Exercise 5

Show that:

$$\log p_\theta(\boldsymbol{x}) \geq \mathrm{E}_{\boldsymbol{Z}\sim q_\phi(\boldsymbol{z}|\boldsymbol{x})}\left[\log p_\theta(\boldsymbol{x} \mid \boldsymbol{Z})\right] - \mathcal{D}_{\mathrm{KL}}\big(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \parallel p(\boldsymbol{z})\big). \tag{20}$$

In this problem, you may use Jensen's inequality without proof: for any convex function ψ,

$$\psi(\mathrm{E}[X]) \leq \mathrm{E}[\psi(X)],$$

where equality holds if and only if X is constant or ψ is linear.

Variational Autoencoder



Variational Autoencoder

- The process of finding the optimal proposal q_{ϕ^*} can be interpreted as maximizing a lower bound of the evidence $\log p_\theta(\boldsymbol{x})$:

$$\log p_\theta(\boldsymbol{x}) \geq \log p_\theta(\boldsymbol{x}) - \mathcal{D}_{\mathrm{KL}}\big(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \parallel p_\theta(\boldsymbol{z} \mid \boldsymbol{x})\big). \tag{21}$$

- Here, the RHS

$$\mathrm{ELBO}(\theta, \phi; \boldsymbol{x}) := \log p_\theta(\boldsymbol{x}) - \mathcal{D}_{\mathrm{KL}}\big(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \parallel p_\theta(\boldsymbol{z} \mid \boldsymbol{x})\big), \tag{22}$$

is referred to as the *evidence lower bound (ELBO)*, which exhibits a computationally tractable form:

$$\mathrm{ELBO}(\theta, \phi; \boldsymbol{x}) = \mathrm{E}_{\boldsymbol{Z} \sim q_\phi(\boldsymbol{z} \mid \boldsymbol{x})}\big[\log p_\theta(\boldsymbol{x} \mid \boldsymbol{Z})\big] - \mathcal{D}_{\mathrm{KL}}\big(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \parallel p(\boldsymbol{z})\big), \tag{23}$$

Variational Autoencoder

Exercise 6

Let $\text{ELBO}(\theta, \phi; \boldsymbol{x}) := \log p_\theta(\boldsymbol{x}) - \mathcal{D}_{\text{KL}}\big(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \parallel p_\theta(\boldsymbol{z} \mid \boldsymbol{x})\big)$. Show that:

$$\text{ELBO}(\theta, \phi; \boldsymbol{x}) = \mathrm{E}_{\boldsymbol{Z} \sim q_\phi(\boldsymbol{z} \mid \boldsymbol{x})}\big[\log p_\theta(\boldsymbol{x} \mid \boldsymbol{Z})\big] - \mathcal{D}_{\text{KL}}\big(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \parallel p(\boldsymbol{z})\big). \quad (24)$$

Variational Autoencoder

- In the ELBO,

$$\text{ELBO}(\theta, \phi; \boldsymbol{x}) = \mathrm{E}_{\boldsymbol{Z} \sim q_\phi(\boldsymbol{z}|\boldsymbol{x})} \left[\log p_\theta(\boldsymbol{x} \mid \boldsymbol{Z}) \right] - \mathcal{D}_{\mathrm{KL}} \left(q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) \parallel p(\boldsymbol{z}) \right), \tag{25}$$

  the first term on the RHS represents the *reconstruction probability*, and maximizing it can be viewed as training a probabilistic autoencoder.

- For example, when $q_\phi(\boldsymbol{z} \mid \boldsymbol{x}) = \mathcal{N}\left(\boldsymbol{z}; \boldsymbol{\mu}_\phi(\boldsymbol{x}), \boldsymbol{\Sigma}_\phi(\boldsymbol{x})\right)$ and $p_\theta(\boldsymbol{x} \mid \boldsymbol{z}) = \mathcal{N}\left(\boldsymbol{x}; \boldsymbol{\mu}_\theta(\boldsymbol{z}), \sigma^2 I_d\right)$, the reconstruction term simplifies to:

$$\mathrm{E}_{\boldsymbol{Z} \sim q_\phi(\boldsymbol{z}|\boldsymbol{x})} \left[\log p_\theta(\boldsymbol{x} \mid \boldsymbol{Z}) \right] = -\frac{1}{2\sigma^2} \mathrm{E}_{\boldsymbol{Z} \sim q_\phi(\boldsymbol{z}|\boldsymbol{x})} \left[\left\| \boldsymbol{x} - \boldsymbol{\mu}_\theta(\boldsymbol{Z}) \right\|^2 \right] - \frac{d}{2} \log(2\pi\sigma^2). \tag{26}$$

- From this perspective, the second term, *KL-penalty term*, acts as a regularizer in the training of autoencoders.

Variational Autoencoder

- The first term can be approximated via the Monte Carlo method:

$$\int_{\mathbb{R}^r} \left(\log p_\theta(\boldsymbol{x}|\boldsymbol{z})\right) q_\phi(\boldsymbol{z}|\boldsymbol{x}) d\boldsymbol{z} \approx M^{-1} \sum_{m=1}^{M} \log p_\theta(\boldsymbol{x}|\boldsymbol{z}^{(m)}) \tag{27}$$

  where $\boldsymbol{z}^{(m)} \overset{\text{i.i.d.}}{\sim} q_\phi(\boldsymbol{z}|\boldsymbol{x})$, and

$$\log p_\theta(\boldsymbol{x}|\boldsymbol{z}^{(m)}) = -\frac{1}{2\sigma^2} \sum_{j=1}^{d} (x_j - (\mu_\theta(\boldsymbol{z}_m))_j)^2 - \frac{d}{2} \log(2\pi\sigma^2). \tag{28}$$

- Under some conditions, this formulation reduces to the reconstruction error of vanilla autoencoders.
- In sampling $\boldsymbol{z}^{(m)} \sim q_\phi(\boldsymbol{z}|\boldsymbol{x}) = \mathcal{N}(\boldsymbol{z}; \boldsymbol{\mu}_\phi(\boldsymbol{x}), \Sigma_\phi(\boldsymbol{x}))$, *reparameterization trick* is applied: $\boldsymbol{Z}^{(m)} = \boldsymbol{\mu}_\phi(\boldsymbol{x}) + \Sigma_\phi(\boldsymbol{x})^{1/2} \boldsymbol{\epsilon}^{(m)}$ where $\boldsymbol{\epsilon}^{(m)} \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, I_r)$.

Variational Autoencoder

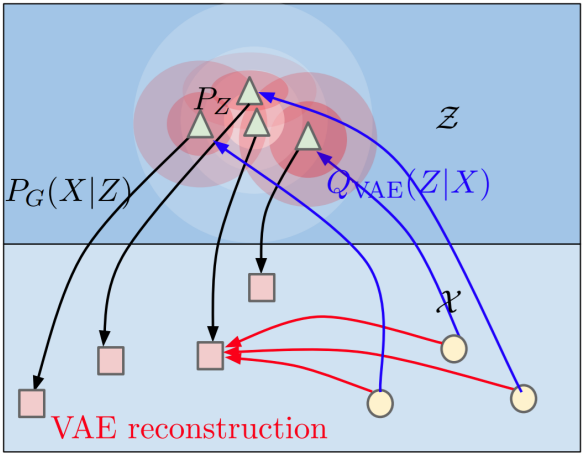
- For the second term, the KL penalty term $\mathcal{D}_{\mathrm{KL}}(q_\phi(\boldsymbol{z}|\boldsymbol{x})||p(\boldsymbol{z}))$, we can use the following formula:

$$\begin{aligned}&\mathcal{D}_{\mathrm{KL}}(\mathcal{N}(\boldsymbol{x};\boldsymbol{\mu}_1,\Sigma_1)||\mathcal{N}(\boldsymbol{x};\boldsymbol{\mu}_2,\Sigma_2))\\&=\frac{1}{2}\left((\boldsymbol{\mu}_2-\boldsymbol{\mu}_1)^T\Sigma_2^{-1}(\boldsymbol{\mu}_2-\boldsymbol{\mu}_1)+\mathrm{tr}(\Sigma_2^{-1}\Sigma_1)-\log\frac{|\Sigma_1|}{|\Sigma_2|}-d\right).\end{aligned} \tag{29}$$

  See StatProofBook contributors (2026) for details on the proof.
- When $p(\boldsymbol{z})=\mathcal{N}(0,I_d)$ and $\Sigma_\phi(\boldsymbol{x})=\mathrm{diag}\left(\sigma_\phi^2(\boldsymbol{x})\right)$, the KL penalty term simplifies to:

$$\mathcal{D}_{\mathrm{KL}}\left(q_\phi(\boldsymbol{z}\mid\boldsymbol{x})\parallel p(\boldsymbol{z})\right)=\frac{1}{2}\sum_{j=1}^{d}\left((\mu_\phi(\boldsymbol{x}))_j^2+(\sigma_\phi^2(\boldsymbol{x}))_j-\log(\sigma_\phi^2(\boldsymbol{x}))_j-1\right). \tag{30}$$

Variational Autoencoder



The figure is from Tolstikhin et al. (2017).

Variational Autoencoder



Images are edited from Kingma and Welling (2014).

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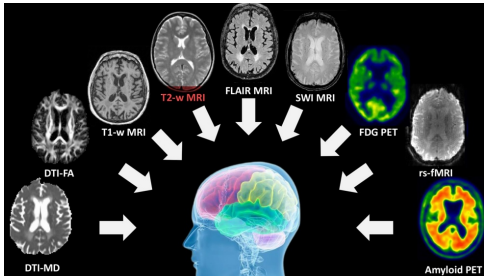
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Motivation



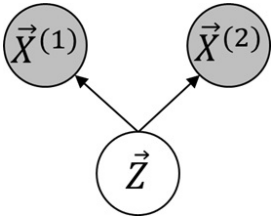
- Data often have multi-modal structures.
- Multi-modalities share comprehensive information while each modality also possesses its unique information.

The figure is from Gong et al. (2023)

Linear Method

- (Traditional) Linear methods are usually based on low-rank matrix compression theory:
  1. Multi-set Canonical Correlation Analysis (Kettenring, 1971)
  2. Joint and Individual Variation Explained (Lock et al., 2013)
  3. Multi-block Partial Least Squares (Chen et al., 2009).
- Multi-modal data may have complicated nonlinear dependencies given latent factors, and we may need to learn mixing mechanisms to analyze new test data.

Joint Multi-modal VAE

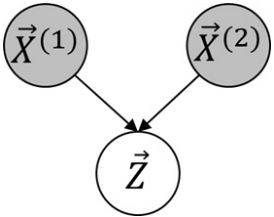


- Joint Multi-modal VAEs (Suzuki et al., 2016) are one of the earliest VAEs-based method, which learn the structure of bi-modal data.
- They assume that two modalities are generated from the same representations:

$$p_\theta(\boldsymbol{z}, \boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}) := p(\boldsymbol{z}) p_\theta(\boldsymbol{x}^{(1)}|\boldsymbol{z}) p_\theta(\boldsymbol{x}^{(2)}|\boldsymbol{z}) \tag{31}$$

where $\boldsymbol{x}^{(1)}$ and $\boldsymbol{x}^{(2)}$ are realizations of the bi-modal data.

Joint Multi-modal VAE



- For the inference, they introduce a joint encoder to approximate the posterior distribution of latent variables given the bi-modal data:

$$q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}). \tag{32}$$

Joint Multi-modal VAE

Exercise 7

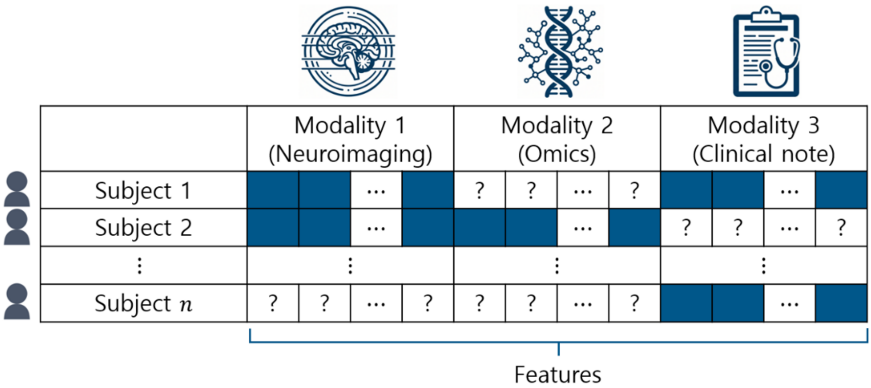
Let $p_\theta(\boldsymbol{z}, \boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}) := p(\boldsymbol{z})p_\theta(\boldsymbol{x}^{(1)}|\boldsymbol{z})p_\theta(\boldsymbol{x}^{(2)}|\boldsymbol{z})$ and let $q_\phi(\boldsymbol{z} \mid \boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)})$ be an inference network. Show that:

$$
\begin{aligned}
& -\log p_\theta(\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}) + \mathcal{D}_{\mathrm{KL}}(q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)})||p_\theta(\boldsymbol{z}|\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)})) \\
&= -\int_{\mathbb{R}^r} \big(\log p_\theta(\boldsymbol{x}^{(1)}|\boldsymbol{z}) + \log p_\theta(\boldsymbol{x}^{(2)}|\boldsymbol{z})\big) q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)}) d\boldsymbol{z} \\
&\quad + \mathcal{D}_{\mathrm{KL}}(q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(2)})||p(\boldsymbol{z}))
\end{aligned} \tag{33}
$$

Joint Multi-modal VAE



Joint Multi-modal VAE



- *Block-missing*, a situation where some modalities are absent, is a chronic problem in multi-modal data analysis (Song et al., 2024).
- Joint Multi-modal VAEs introduce modality-specific encoders to alleviate this issue.

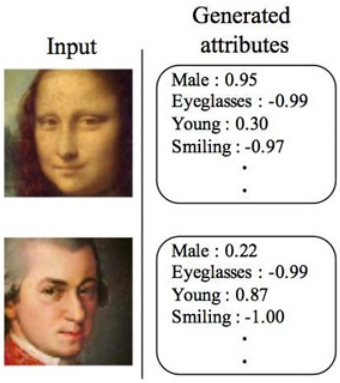
Joint Multi-modal VAE

- Let $q_{\phi^{(1)}}(\boldsymbol{z}|\boldsymbol{x}^{(1)})$ and $q_{\phi^{(2)}}(\boldsymbol{z}|\boldsymbol{x}^{(2)})$ be modality-specific encoders.
- The modified objective function can be expressed as:

$$-\int_{\mathbb{R}^r}\big(\log p_\theta(\boldsymbol{x}^{(1)}|\boldsymbol{z})+\log p_\theta(\boldsymbol{x}^{(2)}|\boldsymbol{z})\big)q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1)},\boldsymbol{x}^{(2)})d\boldsymbol{z}+\mathcal{D}_{\mathrm{KL}}(q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1)},\boldsymbol{x}^{(2)})||p(\boldsymbol{z}))$$
$$+\alpha\big(\mathcal{D}_{\mathrm{KL}}(q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1)},\boldsymbol{x}^{(2)})||q_{\phi^{(1)}}(\boldsymbol{z}|\boldsymbol{x}^{(1)}))+\mathcal{D}_{\mathrm{KL}}(q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1)},\boldsymbol{x}^{(2)})||q_{\phi^{(2)}}(\boldsymbol{z}|\boldsymbol{x}^{(2)}))\big). \tag{34}$$

- Note that all the posterior distributions, $q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1)},\boldsymbol{x}^{(2)})$, $q_{\phi^{(1)}}(\boldsymbol{z}|\boldsymbol{x}^{(1)})$, and $q_{\phi^{(2)}}(\boldsymbol{z}|\boldsymbol{x}^{(2)})$, are modeled as Gaussians, the new added terms have closed-form expressions.

Joint Multi-modal VAE



- Imputation results using $q_{\phi^{(1)}}(\boldsymbol{z}|\boldsymbol{x}^{(1)})$ and $p_\theta(\boldsymbol{x}^{(2)}|\boldsymbol{z})$.

The figure is from Suzuki et al. (2016).

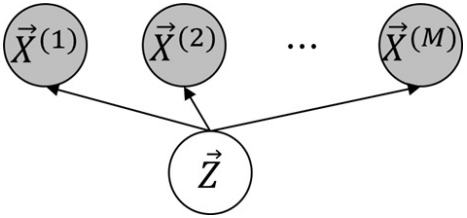
From Bi-modal to Multi-modal

- What happens when the number of modality M increases? When we directly extend the loss function of joint multi-modal VAEs, we have:

$$
\begin{aligned}
&- \log p_\theta(\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(M)}) + \mathcal{D}_{\mathrm{KL}}(q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(M)}) || p_\theta(\boldsymbol{z}|\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(M)})) \\
&= - \int_{\mathbb{R}^r} \Big(\sum_{m=1}^{M} \log p_\theta(\boldsymbol{x}^{(m)}|\boldsymbol{z}) \Big) q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(M)}) d\boldsymbol{z} + \mathcal{D}_{\mathrm{KL}}(q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(M)}) || p(\boldsymbol{z}))
\end{aligned}
\tag{35}
$$

Q: What would be the joint encoder network size for this approach?
Q: Let's consider adding encoders for each observed modality pair. What is the number of possible missing types?

Multi-modal VAE

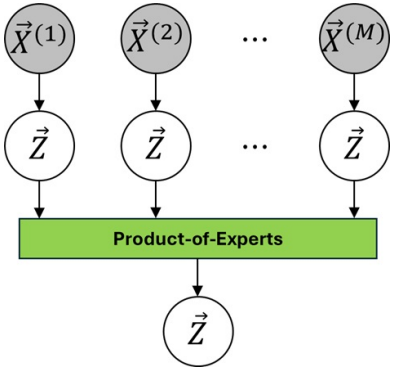


- Multi-modal VAEs (Wu and Goodman, 2018) introduce modality-specific VAEs and aggregate information from each modality using product-of-experts techniques (Hinton, 2002).
- Their generation process can be expressed as:

$$p_{\theta^{(1:M)}}(\boldsymbol{z}, \boldsymbol{x}^{(1:M)}) := p(\boldsymbol{z}) \prod_{m=1}^{M} p_{\theta^{(m)}}(\boldsymbol{x}^{(m)}|\boldsymbol{z}) \tag{36}$$

where $\theta^{(1:M)} := (\theta^{(1)}, \ldots, \theta^{(M)})^T$ and $\boldsymbol{x}^{(1:M)} := (\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(M)})^T$.

Multi-modal VAE



- Product-of-experts techniques enable us to use modality-specific inference networks to build an inference model incorporating all the observed modalities.

Multi-modal VAE

Exercise 8

Let $p_{\theta(1:M)}(\boldsymbol{z}, \boldsymbol{x}^{(1:M)}) := p(\boldsymbol{z}) \prod_{m=1}^{M} p_{\theta(m)}(\boldsymbol{x}^{(m)}|\boldsymbol{z})$. Show that:

$$p_{\theta(1:M)}(\boldsymbol{z}|\boldsymbol{x}^{(1:M)}) \propto \frac{\prod_{m=1}^{M} p_{\theta(m)}(\boldsymbol{z}|\boldsymbol{x}^{(m)})}{p(\boldsymbol{z})^{M-1}} \tag{37}$$

Multi-modal VAE

Exercise 9

Let $p_{\theta(1:M)}(\boldsymbol{z}, \boldsymbol{x}^{(1:M)}) := p(\boldsymbol{z}) \prod_{m=1}^{M} p_{\theta(m)}(\boldsymbol{x}^{(m)}|\boldsymbol{z})$ and let $q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1:M)})$ be an inference network. Show that:

$$
\begin{aligned}
& -\log p_{\theta(1:M)}(\boldsymbol{x}^{(1:M)}) + \mathcal{D}_{\mathrm{KL}}(q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1:M)})||p_{\theta(1:M)}(\boldsymbol{z}|\boldsymbol{x}^{(1:M)})) \\
& = -\int_{\mathbb{R}^r} \Big(\sum_{m=1}^{M} \log p_{\theta(m)}(\boldsymbol{x}^{(m)}|\boldsymbol{z}) \Big) q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1:M)}) d\boldsymbol{z} + \mathcal{D}_{\mathrm{KL}}(q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1:M)})||p(\boldsymbol{z}))
\end{aligned}
\tag{38}
$$

Multi-modal VAE



Multi-modal VAE

- When each modality-specific encoder perfectly approximates the corresponding posterior, i.e., $q_{\phi^{(m)}}(\boldsymbol{z}|\boldsymbol{x}^{(m)}) \approx p_{\theta^{(m)}}(\boldsymbol{z}|\boldsymbol{x}^{(m)})$ for all m, we can utilize them to approximate $p_{\theta^{(1:M)}}(\boldsymbol{z}|\boldsymbol{x}^{(1:M)})$ using Equation (37).
- In implementation, multi-modal VAEs model the quotient $\tilde{q}_{\phi^{(m)}}(\boldsymbol{z}|\boldsymbol{x}^{(m)}) := q_{\phi^{(m)}}(\boldsymbol{z}|\boldsymbol{x}^{(m)})/p(\boldsymbol{z})$ instead of $q_{\phi^{(m)}}(\boldsymbol{z}|\boldsymbol{x}^{(m)})$.
- With quotient networks, the RHS in Equation (37) can be expressed as $p(z)\prod_{m=1}^{M}\tilde{q}_{\phi^{(m)}}(\boldsymbol{z}|\boldsymbol{x}^{(m)})$. All the distributions of prior and quotient networks are Gaussian, implying a closed-expression.

Multi-modal VAE: Loss Function

- The loss function consists of three negative ELBOs.

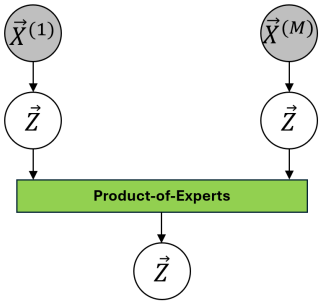
1. $(\boldsymbol{X}^{(1:M)})$ The negative ELBO w.r.t. $\log p_{\theta^{(1:M)}}(\boldsymbol{x}^{(1:M)})$:

$$
\begin{aligned}
&- \log p_{\theta^{(1:M)}}(\boldsymbol{x}^{(1:M)}) + \mathcal{D}_{\mathrm{KL}}(q_{\phi^{(1:M)}}(\boldsymbol{z}|\boldsymbol{x}^{(1:M)})||p_{\theta^{(1:M)}}(\boldsymbol{z}|\boldsymbol{x}^{(1:M)})) \\
&= -\int_{\mathbb{R}^r} \Big(\sum_{m=1}^{M} \log p_{\theta^{(m)}}(\boldsymbol{x}^{(m)}|\boldsymbol{z})\Big) q_{\phi^{(1:M)}}(\boldsymbol{z}|\boldsymbol{x}^{(1:M)}) d\boldsymbol{z} + \mathcal{D}_{\mathrm{KL}}(q_{\phi^{(1:M)}}(\boldsymbol{z}|\boldsymbol{x}^{(1:M)})||p(\boldsymbol{z}))
\end{aligned}
\tag{39}
$$

   where $q_{\phi^{(1:M)}}(\boldsymbol{z}|\boldsymbol{x}^{(1:M)}) \propto p(\boldsymbol{z}) \prod_{m=1}^{M} \tilde{q}_{\phi^{(m)}}(\boldsymbol{z}|\boldsymbol{x}^{(m)})$.

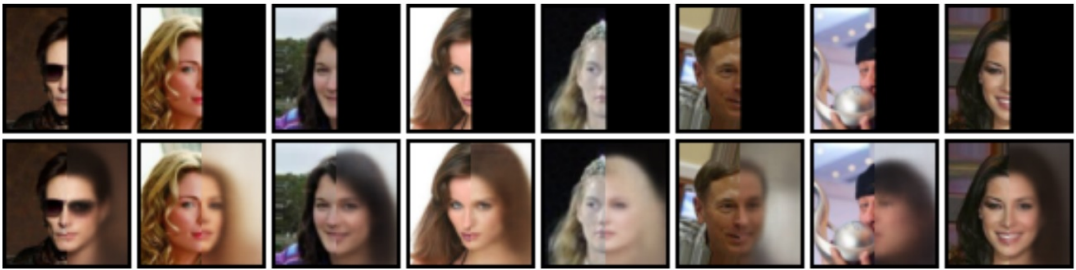
2. $(\boldsymbol{X}^{(m)})$ The negative ELBO w.r.t. $\log p_{\theta^{(m)}}(\boldsymbol{x}^{(m)})$
3. $(\boldsymbol{X}^{(\pi(1):\pi(m))})$ The negative ELBO w.r.t. $\log p_{\theta^{(\pi(1):\pi(m))}}(\boldsymbol{x}^{(\pi(1):\pi(m))})$ where $\pi(1), \ldots, \pi(m)$ are the first m indices from randomly permuted indices.

Multi-modal VAE



- For example, for a sample having only $\boldsymbol{X}^{(1)}$ and $\boldsymbol{X}^{(3)}$, the loss is the summation of the joint ELBO with respect to $(\boldsymbol{x}^{(1)}, \boldsymbol{x}^{(3)})^T$ and the marginal ELBOs with respect to $\boldsymbol{x}^{(1)}$ and $\boldsymbol{x}^{(3)}$.

Multi-modal VAE



The figure is from Wu and Goodman (2018).

Multi-modal VAE

Type	Sentence
$x_{\text{en}} \sim p_{\text{data}}$	this was one of the highest points in my life.
$x_{\text{vi}} \sim p(x_{\text{vi}}\|z(x_{\text{en}}))$	Đó là một gian tôi vời của cuộc đời tôi.
$\text{GOOGLE}(x_{\text{vi}})$	It was a great time of my life.
$x_{\text{en}} \sim p_{\text{data}}$	the project's also made a big difference in the lives of the people .
$x_{\text{vi}} \sim p(x_{\text{vi}}\|z(x_{\text{en}}))$	tôi án này được ra một Điều lớn lao cuộc sống của chúng người sống chữa hưởng .
$\text{GOOGLE}(x_{\text{vi}})$	this project is a great thing for the lives of people who live and thrive .
$x_{\text{vi}} \sim p_{\text{data}}$	trước tiên , tại sao chúng lại có ấn tượng xấu như vậy ?
$x_{\text{en}} \sim p(x_{\text{en}}\|z(x_{\text{vi}}))$	first of all, you do not a good job ?
$\text{GOOGLE}(x_{\text{vi}})$	First, why are they so bad?
$x_{\text{vi}} \sim p_{\text{data}}$	Ông ngoại của tôi là một người thật đáng <unk> phục vào thời ấy .
$x_{\text{en}} \sim p(x_{\text{en}}\|z(x_{\text{vi}}))$	grandfather is the best experience of me family .
$\text{GOOGLE}(x_{\text{vi}})$	My grandfather was a worthy person at the time .

Table 5: Examples of (1) translating English to Vietnamese by sampling from $p(x_{\text{vi}}|z)$ where $z \sim q(z|x_{\text{en}})$, and (2) the inverse. We use Google Translate (GOOGLE) for ground-truth.

The figure is from Wu and Goodman (2018). The training data consist of only 1% completely observed bi-modal data.

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1. Variational Inference

2. Variational Autoencoder

3. Multi-modal Variational Autoencoder

4. Temporal Variational Autoencoder

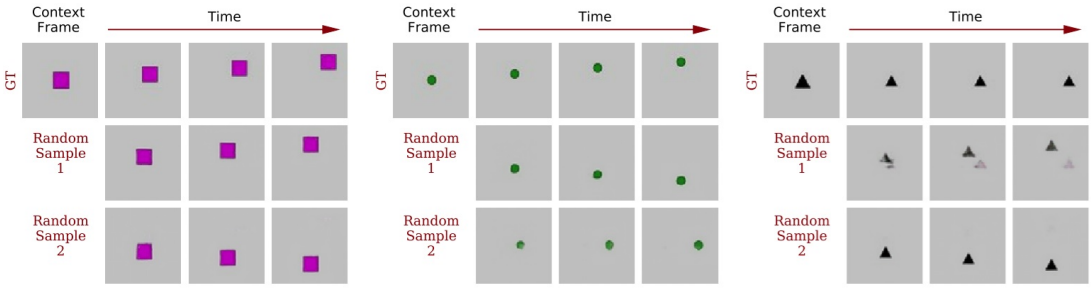
Motivation



- Video frames are popular temporal data, i.e., a time series of image frames at each time step.

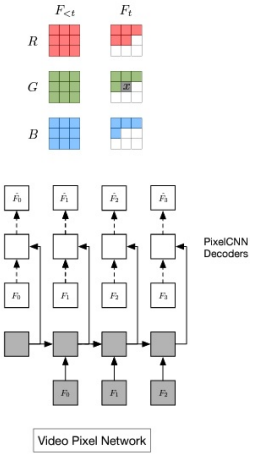
Examples in video generation. Images are from Szeto et al. (2019).

Motivation



Images are edited from Babaeizadeh et al. (2017).

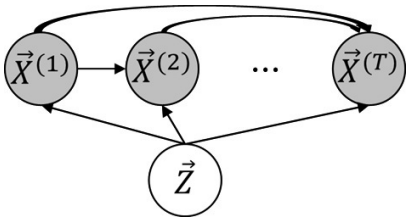
Auto-Regressive Model



- Lotter et al. (2016) proposed a deterministic method that minimizes the MSE between true and predicted future frames.
- This approach can be viewed as an auto-regressive model.
- Video Pixel Network (Kalchbrenner et al., 2017) extended image auto-regressive models to videos.

The figure is from Kalchbrenner et al. (2017).

Stochastic Variational Video Prediction



- Stochastic Variational Video Prediction (SV2P) (Babaeizadeh et al., 2017) is one of the earliest VAE-based video generation methods. It proposes two latent variable models: using (i) time-invariant $\boldsymbol{Z}$ and (ii) time-variant $\boldsymbol{Z}^{(t)}$.
- For (i), the generative model can be expressed as:[3]

$$p_\theta(\boldsymbol{z}, \boldsymbol{x}^{(1:T)}) = p(\boldsymbol{z}) p_\theta(\boldsymbol{x}^{(1)}|\boldsymbol{z}) p_\theta(\boldsymbol{x}^{(2)}|\boldsymbol{z}, \boldsymbol{x}^{(1)}) \cdots p_\theta(\boldsymbol{x}^{(T)}|\boldsymbol{z}, \boldsymbol{x}^{(1:(T-1))}). \quad (40)$$

[3]The SV2P assumes that first few image frames are available at test time.

Stochastic Variational Video Prediction

Exercise 10

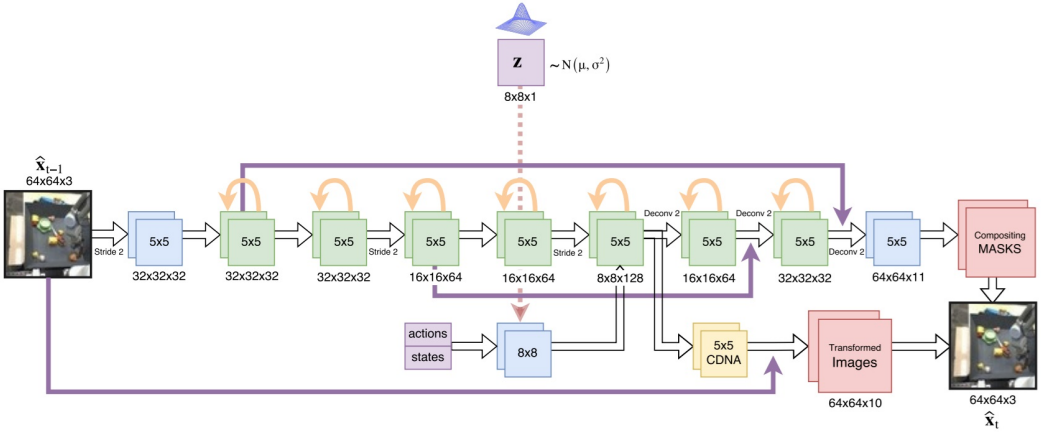
Let $p_\theta(\boldsymbol{z}, \boldsymbol{x}^{(1:T)}) = p(\boldsymbol{z}) p_\theta(\boldsymbol{x}^{(1)}|\boldsymbol{z}) p_\theta(\boldsymbol{x}^{(2)}|\boldsymbol{z}, \boldsymbol{x}^{(1)}) \cdots p_\theta(\boldsymbol{x}^{(T)}|\boldsymbol{z}, \boldsymbol{x}^{(1:(T-1))})$ and let $q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1:T)})$ be an inference network. Show that:

$$
\begin{aligned}
& -\log p_\theta(\boldsymbol{x}^{(1:T)}) + \mathcal{D}_{\mathrm{KL}}(q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1:T)})||p_\theta(\boldsymbol{z}|\boldsymbol{x}^{(1:T)})) \\
&= -\sum_{t=1}^{T} \int_{\mathbb{R}^r} \left(\log p_\theta(\boldsymbol{x}^{(t)}|\boldsymbol{z}, \boldsymbol{x}^{(1:(t-1))}) \right) q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1:T)}) d\boldsymbol{z} \\
&\quad + \mathcal{D}_{\mathrm{KL}}(q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1:T)})||p(\boldsymbol{z})).
\end{aligned} \tag{41}
$$

Stochastic Variational Video Prediction



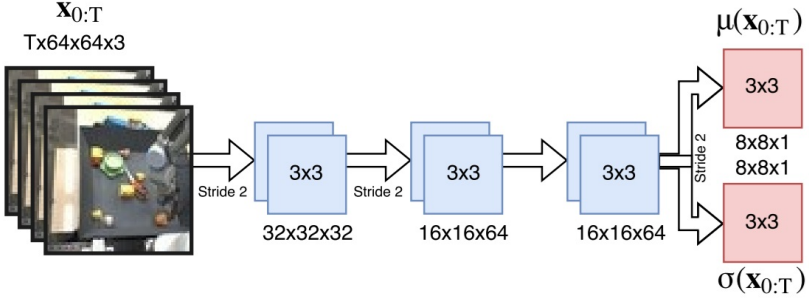
Stochastic Variational Video Prediction



- $p_\theta(\boldsymbol{x}^{(t)}|\boldsymbol{z}, \boldsymbol{x}^{(1:(t-1))})$: The parameter θ is shared across all time steps t.

The figure is edited from Babaeizadeh et al. (2017).

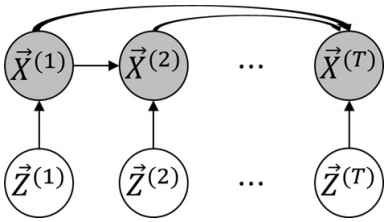
Stochastic Variational Video Prediction



- $q_\phi(\boldsymbol{z}|\boldsymbol{x}^{(1:T)})$: The inference network uses all the image frames to infer the time-invariant latent variable.

The figure is from Babaeizadeh et al. (2017).

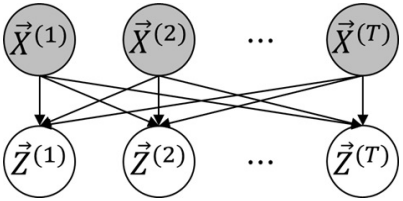
Stochastic Variational Video Prediction



- For the case of (ii) a time-variant latent variable $\boldsymbol{Z}^{(t)}$, the SV2P model formulates the joint distribution as follows:

$$p_\theta(\boldsymbol{z}^{(1:T)}, \boldsymbol{x}^{(1:T)}) = \prod_{t=1}^{T} p(\boldsymbol{z}^{(t)}) \prod_{t=1}^{T} p_\theta(\boldsymbol{x}^{(t)} | \boldsymbol{z}^{(t)}, \boldsymbol{x}^{(1:(t-1))}). \tag{42}$$

Stochastic Variational Video Prediction



- The time-variant $\boldsymbol{Z}^{(t)}$ is inferred using the full sequence of frames $\boldsymbol{X}^{(1:T)}$.
- The inference network can be formalized as

$$q_\phi(\boldsymbol{z}^{(1:T)} \mid \boldsymbol{x}^{(1:T)}) = \prod_{t=1}^{T} q_\phi(\boldsymbol{z}^{(t)} \mid \boldsymbol{x}^{(1:T)}). \tag{43}$$

- Here, the conditional functional form $q_\phi(\cdot \mid \boldsymbol{x}^{(1:T)})$ is assumed to be time-invariant. That is, conditioned on the observed video $\boldsymbol{X}^{(1:T)}$, the latent variables $\boldsymbol{Z}^{(1)}, \ldots, \boldsymbol{Z}^{(T)}$ are conditionally i.i.d.

Stochastic Variational Video Prediction

Exercise 11

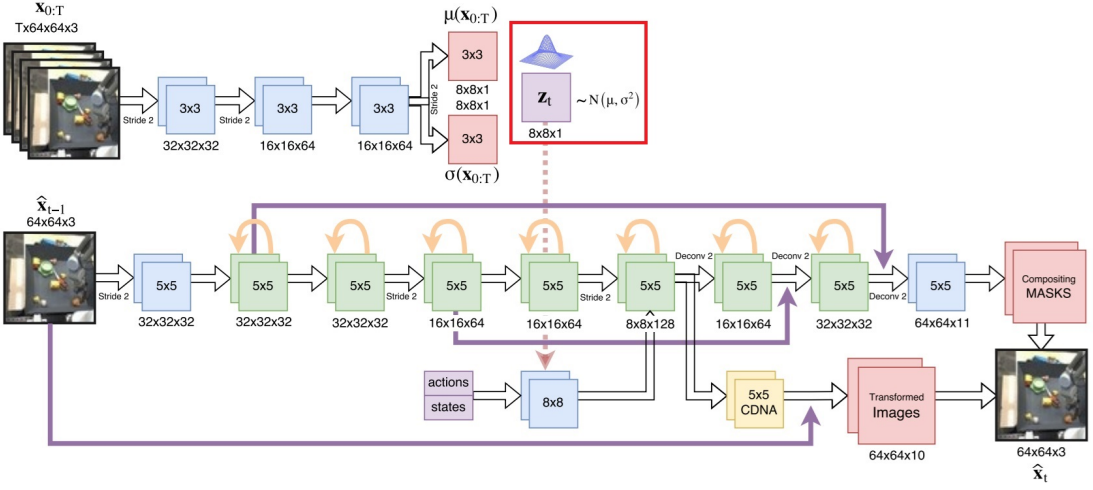
Let $p_\theta(\boldsymbol{z}^{(1:T)}, \boldsymbol{x}^{(1:T)}) = \prod_{t=1}^{T} p(\boldsymbol{z}^{(t)}) \prod_{t=1}^{T} p_\theta(\boldsymbol{x}^{(t)}|\boldsymbol{z}^{(t)}, \boldsymbol{x}^{(1:(t-1))})$ and $q_\phi(\boldsymbol{z}^{(1:T)} \mid \boldsymbol{x}^{(1:T)}) = \prod_{t=1}^{T} q_\phi(\boldsymbol{z}^{(t)} \mid \boldsymbol{x}^{(1:T)})$. Show that:

$$
\begin{aligned}
& -\log p_\theta(\boldsymbol{x}^{(1:T)}) + \mathcal{D}_{\mathrm{KL}}(q_\phi(\boldsymbol{z}^{(1:T)}|\boldsymbol{x}^{(1:T)})||p_\theta(\boldsymbol{z}^{(1:T)}|\boldsymbol{x}^{(1:T)})) \\
&= \sum_{t=1}^{T} \Big(-\int_{\mathbb{R}^r} \big(\log p_\theta(\boldsymbol{x}^{(t)}|\boldsymbol{z}^{(t)}, \boldsymbol{x}^{(1:(t-1))}) \big) q_\phi(\boldsymbol{z}^{(t)}|\boldsymbol{x}^{(1:T)}) d\boldsymbol{z}^{(t)} \\
&\quad + \mathcal{D}_{\mathrm{KL}}(q_\phi(\boldsymbol{z}^{(t)}|\boldsymbol{x}^{(1:T)})||p(\boldsymbol{z}^{(t)})) \Big)
\end{aligned}
\tag{44}
$$

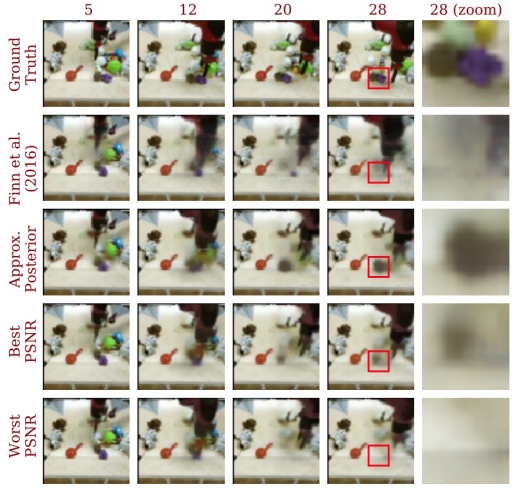
Stochastic Variational Video Prediction



Stochastic Variational Video Prediction

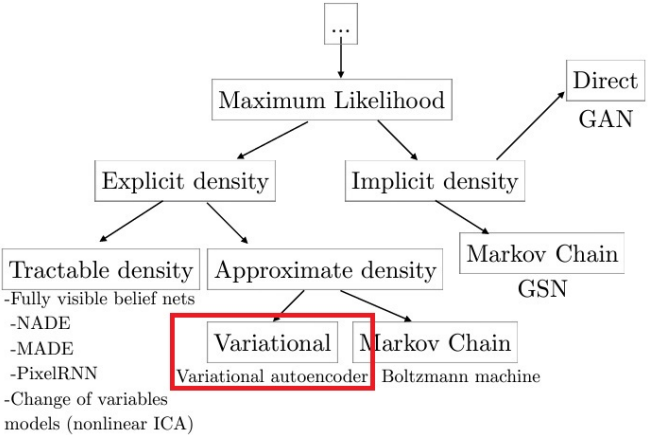


Stochastic Variational Video Prediction



Images are edited from Babaeizadeh et al. (2017). PSNR stands for Peak Signal-to-Noise Ratio.

Lecture Topics



The figure is edited from Goodfellow (2016).

References I

Babaeizadeh, M., Finn, C., Erhan, D., Campbell, R. H., and Levine, S. (2017). Stochastic variational video prediction. *arXiv preprint arXiv:1710.11252*.

Bishop, C. M. (2006). Pattern recognition and machine learning. *Springer google schola*, 2:1122–1128.

Chen, K., Reiman, E. M., Huan, Z., Caselli, R. J., Bandy, D., Ayutyanont, N., and Alexander, G. E. (2009). Linking functional and structural brain images with multivariate network analyses: a novel application of the partial least square method. *Neuroimage*, 47(2):602–610.

Gong, C., Jing, C., Chen, X., Pun, C. M., Huang, G., Saha, A., Nieuwoudt, M., Li, H.-X., Hu, Y., and Wang, S. (2023). Generative ai for brain image computing and brain network computing: a review. *Frontiers in Neuroscience*, 17:1203104.

Goodfellow, I. (2016). Nips 2016 tutorial: Generative adversarial networks. *arXiv preprint arXiv:1701.00160*.

References II

Hinton, G. E. (2002). Training products of experts by minimizing contrastive divergence. *Neural computation*, 14(8):1771–1800.

Kalchbrenner, N., Oord, A., Simonyan, K., Danihelka, I., Vinyals, O., Graves, A., and Kavukcuoglu, K. (2017). Video pixel networks. In *International Conference on Machine Learning*, pages 1771–1779. PMLR.

Kettenring, J. R. (1971). Canonical analysis of several sets of variables. *Biometrika*, 58(3):433–451.

Kim, J.-H., Zhang, Y., Han, K., Wen, Z., Choi, M., and Liu, Z. (2021). Representation learning of resting state fmri with variational autoencoder. *NeuroImage*, 241:118423.

Kingma, D. P. and Welling, M. (2014). Auto-encoding variational bayes. In *Proceedings of the International Conference on Learning Representations (ICLR)*.

Kingma, D. P., Welling, M., et al. (2019). An introduction to variational autoencoders. *Foundations and Trends® in Machine Learning*, 12(4):307–392.

References III

Lock, E. F., Hoadley, K. A., Marron, J. S., and Nobel, A. B. (2013). Joint and individual variation explained (jive) for integrated analysis of multiple data types. *The annals of applied statistics*, 7(1):523.

Lotter, W., Kreiman, G., and Cox, D. (2016). Deep predictive coding networks for video prediction and unsupervised learning. *arXiv preprint arXiv:1605.08104*.

Song, S., Lin, Y., and Zhou, Y. (2024). Semi-supervised inference for block-wise missing data without imputation. *Journal of Machine Learning Research*, 25(99):1–36.

StatProofBook contributors (2026). Kullback-leibler divergence between multivariate normal distributions. `https://statproofbook.github.io/P/mvn-kl.html`. Accessed: 2026-06-08.

Suzuki, M., Nakayama, K., and Matsuo, Y. (2016). Joint multimodal learning with deep generative models. *arXiv preprint arXiv:1611.01891*.

References IV

Szeto, R., Sun, X., Lu, K., and Corso, J. J. (2019). A temporally-aware interpolation network for video frame inpainting. *IEEE transactions on pattern analysis and machine intelligence*, 42(5):1053–1068.

Tolstikhin, I., Bousquet, O., Gelly, S., and Schoelkopf, B. (2017). Wasserstein auto-encoders. *arXiv preprint arXiv:1711.01558*.

Wu, M. and Goodman, N. (2018). Multimodal generative models for scalable weakly-supervised learning. *Advances in neural information processing systems*, 31.